CV5600 · Geospatial Data Science · IIT Madras

HYDRO GRAPH

A Spatiotemporal Graph Neural Network for Hyper-Local Urban Flood Forecasting

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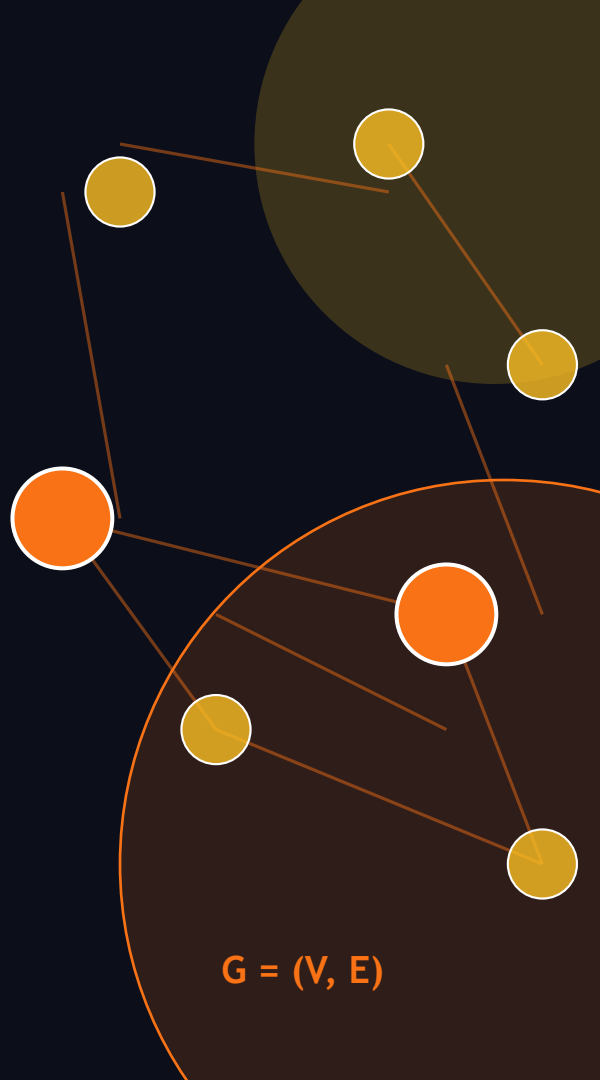


MOTIVATION

The Urban Flood Crisis

Floods are now India's #1 natural disaster

₹1.8 Lakh Cr

Total flood damage in India (2000-2022)

500+

Lives lost in Chennai floods (2015)

46%

Chennai urban area under flood risk

72 hrs

Warning window needed to evacuate

3×

Flood frequency rise by 2050 (IPCC)

10m

Sentinel-2 spatial resolution we exploit

Study Area: Chennai Metropolitan Region — India's most flood-vulnerable major city

The Wrong Question

CNN and Random Forest models ask:

"Is this pixel impervious surface?"

But water doesn't care about pixels.

It flows along topology — roads, drains, river channels, catchment boundaries.

Formal Goal

Learn $f : G(V,E), \{X_v\}, \{R(t)\} \rightarrow \hat{y}_v(t+\Delta)$
for all $v \in V$

Spatial Independence Assumption

Each pixel treated as isolated. No modelling of how water propagates from Node A → Node B through connected drains.

Temporal Blindness

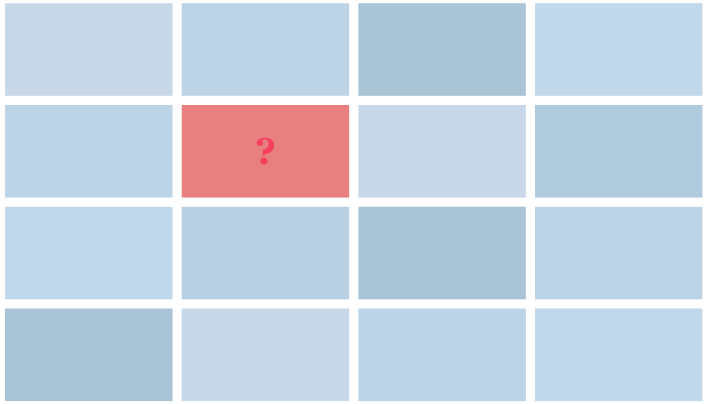
Static snapshots ignore rainfall lag — runoff from upstream arrives 30-90 min after rainfall peak. Temporal GRU captures this.

Scaling Bottleneck

Image-based models waste memory processing millions of irrelevant pixels. Traditional graph models crash trying to load an entire city network at once.

"Water doesn't care about pixels. It flows along topology."

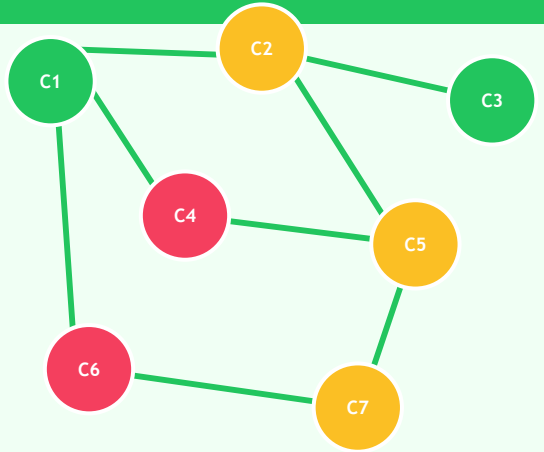
Traditional: Grid / Image Model



✗ Is this pixel impervious?
✗ No flow direction
✗ No network connectivity

VS

Hydro-Graph: G = (V, E)



✓ Flow-aware topology
✓ Catchment connectivity
✓ Hydrologically grounded

PROPOSED METHODOLOGY · ST-GNN ARCHITECTURE

01
Graph Construction

Sentinel-2 + OSM
→ G=(V,E)

OSMnx extracts road/drain topology; Sentinel-2 defines node boundaries

02
Physics Features

DEM + LULC
+ Flow Accum.

Slope, elevation, imperviousness, TWI per node; physically grounded

03
Temporal Encoding

Rainfall t-series
→ GRU embed.

Per-node 6-hr rainfall history; GRU hidden state captures lag effect

04
ST-GNN Layers

GraphSAGE
+ GRU fusion

Spatial: message passing over G; Temporal: GRU across time-steps

05
Flood Risk Output

$\hat{y}_v(t+\Delta)$
per node

Node-level binary & probabilistic flood susceptibility + risk heat-map

PyTorch Geometric | GraphSAGE / GAT | GRU / LSTM | Google Earth Engine | OSMnx 1.x | scikit-learn

FEATURE ENGINEERING · PHYSICS-AWARE NODE ATTRIBUTES

SPATIAL / TERRAIN

Elevation (Z)
SRTM 30m DEM

via ALOS/SRTM

Slope (β)
∇Z per catchment

deg / $m \cdot m^{-1}$

TWI
Topographic Wetness

ln(A / tan β)

Flow Accumulation
D8 Algorithm

upstream area

Catchment Area
Watershed boundary

Km2

Drainage Infrastructure Capacity
Underground Z-Axis

m

SPECTRAL / LULC

NDVI
Vegetation density

(NIR−R) / (NIR+R)

NDWI
Water body index

(G−NIR) / (G+NIR)

NDBI
Built-up intensity

(SWIR−NIR) / (…)

Imperviousness
% sealed surface

f(NDBI, NDVI)

Soil Moisture
SAR C-band proxy

Sentinel-1 VV

TEMPORAL / RAINFALL

R(t)
Hourly precipitation

GPM IMERG 0.1°

R(t−1..−6)
6-hr lag window

GRU Input seq.

Tidal Data

GRU input seq.

Antecedent SM
Soil saturation

5-day precip.

Rainfall Intensity
mm/hr peak

IMD gauge data

Storm Duration
Event length

onset/cessation

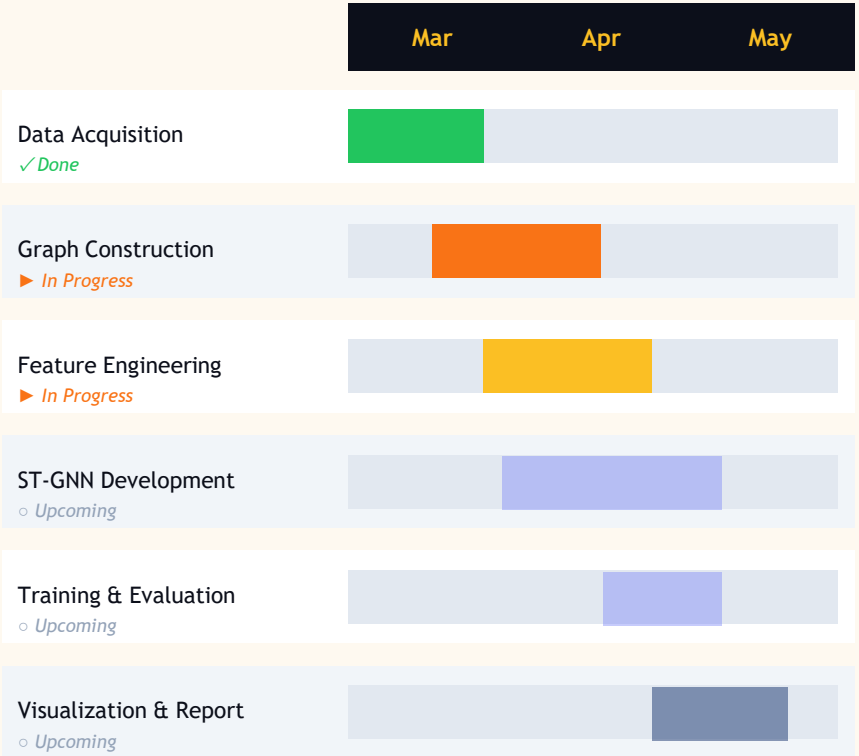
DATA SOURCES · ALL OPEN ACCESS

Dataset	Provider	Resolution	Key Variables	Access
Sentinel-2 MSI	ESA Copernicus	10 m optical	NDVI, NDWI, NDBI, RGB bands (B2–B12)	Free / GEE
OpenStreetMap	OSM Foundation	Vector / ~1m	Road network, drains, LULC polygons	Open / OSMnx
SRTM / ALOS DEM	NASA / JAXA	30 m raster	Elevation, slope, aspect, TWI, FAC	Free / GEE
GPM IMERG	NASA	0.1° × 30 min	Precipitation rate, 20-yr historical	Free / GEE
IMD Gauge Data	India Met. Dept.	Station-level	Hourly rainfall, ground truth validation	Request / API
Flood Archive (DFO)	Dartmouth / GFM	Event-level	Historical flood extents for training labels	Open / CSV

Completed

- Study area formalized: Chennai Metropolitan (flood-prone, 2015 event ground truth available)
- Problem re-framed: from pixel classification → graph-structured spatiotemporal ML
- Literature survey: GNNs for hydrology (STGCN, DCRNN, GraphSAGE in flood prediction)
- All data sources identified + access confirmed: Sentinel-2, OSM, SRTM, GPM, DFO archive
- GEE preprocessing pipeline designed; OSMnx graph extraction prototype planned

Timeline



HYDRO-GRAPH · INITIAL PRESENTATION · CV5600

A New Lens for Urban Floods

Not pixels. Networks.

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Thank You